

Theory and Numerics of a Boundary-Element Fluid–Structure Solver for Multiple Ellipsoids in Potential Flow

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Abstract

This document sets out the complete mathematical and numerical formulation of `multi_ellip`, a solver for the coupled motion of N rigid ellipsoidal bodies immersed in an unbounded, inviscid, incompressible and initially irrotational fluid. The flow is governed by Laplace’s equation for a velocity potential, solved at each instant by a boundary–element method (BEM) on six-node curvilinear triangular meshes. Hydrodynamic loads are obtained from *Lamb’s hydrodynamic impulse* rather than from a direct pressure integral, which makes the discrete fluid–structure coupling a state function of the body configuration and velocities and therefore (to integration tolerance) momentum- and energy-consistent. The bodies obey the Newton–Euler equations, integrated with a quaternion exponential-map scheme for the rotational degrees of freedom and an implicit-midpoint impulse scheme for the coupled translation. We give the governing equations, the boundary-integral formulation and its discretisation, the analytic ellipsoid added-mass tensors, the impulse-based load model and its conservation properties, and the coupled time-integration algorithm together with the linear-algebra and convergence-acceleration techniques used to make it efficient.

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1 Problem statement

We consider N rigid bodies $B_1, \dots, B_N$, each an ellipsoid with semi-axes (a_i, b_i, c_i) , moving through an unbounded fluid domain $\Omega = \mathbb{R}^3 \setminus \bigcup_i \overline{B_i}$. Body i has mass m_i , centre of mass $\mathbf{c}_i(t)$, lab-frame translational velocity $\mathbf{v}_i(t)$, and lab-frame angular velocity $\boldsymbol{\omega}_i(t)$; its orientation is carried by a unit quaternion $q_i(t)$. The fluid has density ρ_f and the solid density of body i is $\rho_{s,i}$.

The fluid is inviscid, incompressible and the flow is irrotational, so the velocity field admits a potential,

$$\mathbf{u}(\mathbf{x}, t) = \nabla \phi(\mathbf{x}, t), \quad \mathbf{x} \in \Omega. \quad (1)$$

Incompressibility $\nabla \cdot \mathbf{u} = 0$ then makes ϕ harmonic,

$$\nabla^2 \phi = 0 \quad \text{in } \Omega. \quad (2)$$

The flow is driven entirely by the moving boundaries. On the surface $S_i = \partial B_i$ the no-penetration condition equates the normal fluid velocity to the normal velocity of the rigid surface,

$$\partial_n \phi(\mathbf{x}) \equiv \hat{\mathbf{n}}(\mathbf{x}) \cdot \nabla \phi = \hat{\mathbf{n}}(\mathbf{x}) \cdot [\mathbf{v}_i + \boldsymbol{\omega}_i \times (\mathbf{x} - \mathbf{c}_i)], \quad \mathbf{x} \in S_i, \quad (3)$$

with $\hat{\mathbf{n}}$ the unit normal pointing into the fluid. The fluid is at rest at infinity, $\nabla \phi \rightarrow \mathbf{0}$ as $|\mathbf{x}| \rightarrow \infty$. Because eq. (2)–eq. (3) contain no time derivative of ϕ , the potential is determined *instantaneously* by the current configuration $\{\mathbf{c}_i, q_i\}$ and velocities $\{\mathbf{v}_i, \boldsymbol{\omega}_i\}$: the fluid carries no independent state. This is the structural fact the solver exploits throughout.

The unknowns of the coupled problem are the rigid-body trajectories. The fluid enters only through the loads it exerts, which we compute from ϕ .

2 Boundary-integral formulation

2.1 Representation and the second-kind equation

Let $G(\mathbf{x}, \mathbf{x}_0) = \frac{1}{4\pi |\mathbf{x} - \mathbf{x}_0|}$ be the free-space Green's function of the Laplacian, $\nabla^2 G = -\delta(\mathbf{x} - \mathbf{x}_0)$. Applying Green's third identity to the harmonic ϕ in Ω and taking the field point $\mathbf{x}_0$ to

the (smooth) boundary yields a boundary-integral equation relating the surface potential ϕ to its normal derivative $\partial_n \phi$:

$$-\frac{1}{2}\phi(\mathbf{x}_0) + \underbrace{\text{p.v.} \int_S \phi(\mathbf{x}) \frac{\partial G(\mathbf{x}, \mathbf{x}_0)}{\partial n(\mathbf{x})} dS(\mathbf{x})}_{\text{double-layer } (\mathcal{D}\phi)(\mathbf{x}_0)} = \underbrace{\int_S G(\mathbf{x}, \mathbf{x}_0) \partial_n \phi(\mathbf{x}) dS(\mathbf{x})}_{\text{single-layer } (\mathcal{S}\partial_n \phi)(\mathbf{x}_0)}, \quad \mathbf{x}_0 \in S, \quad (4)$$

where $S = \bigcup_i S_i$ and p.v. denotes the principal-value double-layer integral. The $-\frac{1}{2}$ term is the jump relation of the double-layer potential across the smooth surface. Equation (3) fixes the right-hand side, so eq. (4) is a Fredholm equation of the *second kind* for the unknown surface potential ϕ ,

$$(\mathcal{D} - \tfrac{1}{2}\mathcal{I})\phi = \mathcal{S}\partial_n \phi. \quad (5)$$

Second-kind equations are well conditioned, which is central to the fast iterative solution in section 7.

2.2 Surface discretisation

Each ellipsoid surface is meshed by recursive subdivision of a base octahedron. Starting from 8 triangular faces, n_{div} levels of $1 \rightarrow 4$ subdivision give $8 \cdot 4^{n_{\text{div}}}$ flat triangles whose vertices and edge midpoints are projected onto the unit sphere and then scaled to the ellipsoid semi-axes and rotated/translated into place. Each triangle is a *six-node curvilinear element* carrying three vertex nodes and three edge-midpoint nodes; duplicate nodes are merged to give a watertight mesh with $n_{\text{pts}} = 2(8 \cdot 4^{n_{\text{div}}}) + 2$ global nodes per body.

On the reference triangle with barycentric-like parameters (ξ, η) the field is interpolated with six quadratic isoparametric shape functions $\{\varphi_m(\xi, \eta)\}_{m=1}^6$, curvature-corrected by per-element parameters (α, β, γ) that locate the edge nodes,

$$\mathbf{x}(\xi, \eta) = \sum_{m=1}^6 \varphi_m(\xi, \eta) \mathbf{x}_m, \quad \phi(\xi, \eta) = \sum_{m=1}^6 \varphi_m(\xi, \eta) \phi_m. \quad (6)$$

The surface metric follows from the tangent vectors $\mathbf{x}_{,\xi} = \partial_\xi \mathbf{x}$, $\mathbf{x}_{,\eta} = \partial_\eta \mathbf{x}$: the (unnormalised) normal is $\mathbf{x}_{,\xi} \times \mathbf{x}_{,\eta}$ and the area element is $dS = h_s d\xi d\eta$ with $h_s = \|\mathbf{x}_{,\xi} \times \mathbf{x}_{,\eta}\|$. Nodal normals are obtained by averaging element normals at shared nodes.

2.3 Collocation system

Collocating eq. (4) at every global node $\mathbf{x}_0 = \mathbf{x}_i$ produces a dense linear system for the nodal potentials $\boldsymbol{\phi} = (\phi_1, \dots, \phi_{n_{\text{pts}}})^\top$,

$$\mathbf{A} \boldsymbol{\phi} = \mathbf{b}, \quad \mathbf{A} = \mathbf{D} - \tfrac{1}{2}\mathbf{I}, \quad \mathbf{b} = \mathbf{S} \mathbf{q}, \quad (7)$$

where $\mathbf{q}$ is the vector of nodal Neumann data $\partial_n \phi$ from eq. (3), $\mathbf{D}$ is the discrete double-layer operator and $\mathbf{S}$ the discrete single-layer operator. Element integrals are evaluated by Gauss-triangle quadrature with m_{int} points for regular (non-singular) pairs. When the collocation node coincides with a node of the integration element the integrand is singular; the element is subdivided about the singular node and each sub-triangle is integrated with a Gauss-Legendre rule of order n_q in a local polar parametrisation that absorbs the $1/|\mathbf{x} - \mathbf{x}_0|$ singularity. The discrete entry on the diagonal carries the analytic $-\frac{1}{2}$ jump term of eq. (4).

A key invariance is used repeatedly below: the operators $\mathbf{D}$ and $\mathbf{S}$ depend only on the *geometry* of the surfaces (positions and orientations), not on the body velocities. Moreover, for a single rigid body the influence matrix $\mathbf{A}$ is invariant under rigid-body motion (rotation and translation of the mesh leave the kernels unchanged in the body frame), so it need be assembled and factorised only once.

3 Hydrodynamic loads from Lamb's impulse

3.1 Impulse, kinetic energy, and why we avoid the pressure integral

The hydrodynamic force and torque could in principle be obtained from the unsteady Bernoulli pressure $p = -\rho_f(\partial_t\phi + \frac{1}{2}|\nabla\phi|^2)$ integrated over each surface. That route requires the partial time derivative $\partial_t\phi$, which on a moving mesh is awkward and, when differenced in time, injects a spurious high-frequency mode that manifests as an energy oscillation. Instead we use *Lamb's hydrodynamic impulse*—the impulsive force/torque that would generate the present flow from rest. For body i it is the surface integral of the potential weighted by the normal,

$$\mathbf{L}_i = \rho_f \oint_{S_i} \phi \hat{\mathbf{n}} dS, \quad (\text{linear impulse}) \quad (8)$$

$$\mathbf{\Lambda}_i = \rho_f \oint_{S_i} \phi (\mathbf{r} \times \hat{\mathbf{n}}) dS, \quad \mathbf{r} = \mathbf{x} - \mathbf{c}_i, \quad (\text{angular impulse about } \mathbf{c}_i). \quad (9)$$

The angular impulse eq. (9) is taken about the *moving body centre* $\mathbf{c}_i$; this choice is important for the transport term derived below. The fluid kinetic energy is the boundary integral

$$T_f = \frac{1}{2}\rho_f \int_{\Omega} |\nabla\phi|^2 dV = \frac{1}{2}\rho_f \oint_S \phi \partial_n \phi dS, \quad (10)$$

where the volume integral collapses to the surface by the divergence theorem and harmonicity. Crucially, $\mathbf{L}_i$, $\mathbf{\Lambda}_i$ and T_f are *state functions*: linear functionals of ϕ , which is itself a linear function of the body velocities at fixed geometry. Differencing them between time levels therefore gives a discretely consistent load.

3.2 Force, torque, and the reference-point transport term

For a single body in an unbounded ideal fluid the total (fluid + solid) momentum is conserved, and the force exerted by the fluid equals the rate of change of the fluid linear impulse,

$$\mathbf{F}_i = \frac{d\mathbf{L}_i}{dt}. \quad (11)$$

The body momentum then obeys $m_i \dot{\mathbf{v}}_i = \mathbf{F}_i = \dot{\mathbf{L}}_i$, i.e. the combination $m_i \mathbf{v}_i - \mathbf{L}_i$ is conserved (the effective inertia is the solid mass plus the added mass, section 4).

The torque requires care because $\mathbf{\Lambda}_i$ is referred to the translating point $\mathbf{c}_i$. Conservation of the *total* angular momentum of the fluid–solid system about a fixed lab origin O is the physical constraint. The angular momentum about O contributed by body i is the solid spin $\mathbf{h}_i = \mathbf{R}_i \mathbf{l}_i^{\text{body}} \boldsymbol{\omega}_i^{\text{body}}$ plus the orbital term, against which the fluid impulse about O enters with the sign fixed by the momentum convention. Writing the conserved per-body invariants

$$\mathbf{p}_i = m_i \mathbf{v}_i - \mathbf{L}_i, \quad \mathbf{H}_i = \mathbf{c}_i \times \mathbf{p}_i + \mathbf{h}_i - \mathbf{\Lambda}_i, \quad (12)$$

and using the shift of reference point $\mathbf{\Lambda}_i^O = \mathbf{\Lambda}_i + \mathbf{c}_i \times \mathbf{L}_i$ (moving the moment from $\mathbf{c}_i$ to O), one finds that $d\mathbf{H}_i/dt = 0$ requires the torque about the body centre to be

$$\boxed{\mathbf{N}_i = \frac{d\mathbf{\Lambda}_i}{dt} - \mathbf{v}_i \times \mathbf{p}_i} \quad (13)$$

where $\dot{\mathbf{c}}_i = \mathbf{v}_i$ and, in the discrete scheme, the linear update enforces $\dot{\mathbf{p}}_i = 0$ per body. The second term $-\mathbf{v}_i \times \mathbf{p}_i$ is the *reference-point transport term*: it accounts for the angular impulse being measured about a translating origin. Omitting it leaves the discrete torque inconsistent and lets the total angular momentum drift at the rate $\sum_i \mathbf{v}_i \times \mathbf{p}_i$, which pumps energy into the rotational degrees of freedom—the dominant error source at low body-to-fluid density ratio. The transport term in eq. (13) holds for any number of bodies because the linear update conserves $\mathbf{p}_i$ for each body individually.

3.3 Discrete conserved quantities

Summed over bodies, the scheme conserves (to integration tolerance) the total linear and angular momentum

$$\mathbf{P} = \sum_i (m_i \mathbf{v}_i - \mathbf{L}_i), \quad \mathbf{H} = \sum_i \left(\mathbf{c}_i \times (m_i \mathbf{v}_i - \mathbf{L}_i) + \mathbf{h}_i - \mathbf{\Lambda}_i \right), \quad (14)$$

and the total kinetic energy $T = T_f + T_s$ with the solid part $T_s = \sum_i \left(\frac{1}{2} m_i |\mathbf{v}_i|^2 + \frac{1}{2} \boldsymbol{\omega}_i^{\text{body}\top} \mathbf{l}_i^{\text{body}} \boldsymbol{\omega}_i^{\text{body}} \right)$. These quantities are monitored online and are the primary verification diagnostics.

4 Analytic ellipsoid added mass

For an isolated ellipsoid the added-mass tensors are known in closed form (Lamb, Kirchhoff). Define the Lamb shape factors by the elliptic integrals

$$\alpha_0 = abc \int_0^\infty \frac{d\lambda}{(a^2 + \lambda) \Delta(\lambda)}, \quad \beta_0 = abc \int_0^\infty \frac{d\lambda}{(b^2 + \lambda) \Delta(\lambda)}, \quad \gamma_0 = abc \int_0^\infty \frac{d\lambda}{(c^2 + \lambda) \Delta(\lambda)}, \quad (15)$$

with $\Delta(\lambda) = \sqrt{(a^2 + \lambda)(b^2 + \lambda)(c^2 + \lambda)}$ and $\alpha_0 + \beta_0 + \gamma_0 = 2$. These are computed by numerically integrating the equivalent ODE system $d\alpha_0/d\lambda$, etc., with an adaptive Dormand–Prince (DOP853) integrator. The body-frame added-mass tensor is diagonal,

$$\mathbf{M}_f = \rho_f V_s \text{diag} \left(\frac{\alpha_0}{2 - \alpha_0}, \frac{\beta_0}{2 - \beta_0}, \frac{\gamma_0}{2 - \gamma_0} \right), \quad V_s = \frac{4}{3} \pi abc, \quad (16)$$

and the added moment of inertia $\mathbf{l}_f$ is given by the corresponding Lamb expressions in terms of $(\alpha_0, \beta_0, \gamma_0)$ and the semi-axes. The solid mass and inertia are

$$m_s = \rho_s V_s, \quad \mathbf{l}_s = \frac{m_s}{5} \text{diag}(b^2 + c^2, a^2 + c^2, a^2 + b^2). \quad (17)$$

The added-mass tensor eq. (16) is exact only for an isolated body; with several nearby bodies the true added mass acquires off-diagonal, configuration-dependent couplings that the BEM resolves implicitly. The analytic $\mathbf{M}_f$ is therefore used not as the load model but as a *preconditioner* (section 6): the effective body-frame inertia $\mathbf{P}_i = m_i \mathbf{l} + \mathbf{M}_{f,i}$ captures the dominant self-added-mass and makes the implicit velocity solve well conditioned even when $\rho_s/\rho_f \ll 1$.

5 Rigid-body dynamics and rotational integration

Each body obeys the Newton–Euler equations. Translation is $m_i \dot{\mathbf{v}}_i = \mathbf{F}_i$ with the impulse force eq. (11). Rotation is written in the body frame, where the inertia tensor is constant, as Euler’s equation

$$\mathbf{l}_i^{\text{body}} \dot{\boldsymbol{\omega}}_i^{\text{body}} + \boldsymbol{\omega}_i^{\text{body}} \times (\mathbf{l}_i^{\text{body}} \boldsymbol{\omega}_i^{\text{body}}) = \mathbf{N}_i^{\text{body}}, \quad (18)$$

with $\mathbf{N}_i$ from eq. (13) rotated into the body frame. Lab and body frames are related by the unit quaternion q_i through the sandwich product: for a pure (vector) quaternion $\mathbf{p}$, $\mathbf{p}^{\text{lab}} = q \mathbf{p}^{\text{body}} q^{-1}$ and $\mathbf{p}^{\text{body}} = q^{-1} \mathbf{p}^{\text{lab}} q$.

5.1 Quaternion exponential-map update

Orientation is advanced by the exponential map, which keeps q exactly on the unit sphere (no renormalisation drift). For a lab-frame angular velocity $\boldsymbol{\omega}$ held over a step Δt ,

$$q_{n+1} = \exp \left(\frac{1}{2} \boldsymbol{\omega} \Delta t \right) q_n, \quad \exp \left(\frac{1}{2} \boldsymbol{\omega} \Delta t \right) = \left(\cos \frac{|\boldsymbol{\omega}| \Delta t}{2}, \frac{\sin \frac{|\boldsymbol{\omega}| \Delta t}{2}}{|\boldsymbol{\omega}|} \boldsymbol{\omega} \right), \quad (19)$$

a rotation by angle $|\boldsymbol{\omega}| \Delta t$ about the instantaneous axis $\boldsymbol{\omega}/|\boldsymbol{\omega}|$. The angular velocity itself is advanced by Euler’s equation eq. (18).

5.2 Predictor–corrector direction method (PCDM)

The rotational state is integrated by a second-order predictor–corrector. With the step-averaged torque $\mathbf{N}$ held fixed, a half step from $(q_n, \boldsymbol{\omega}_n)$ predicts the midpoint orientation and angular velocity; the torque and Euler right-hand side are then re-evaluated at the midpoint and a full step advances $(q_n, \boldsymbol{\omega}_n) \rightarrow (q_{n+1}, \boldsymbol{\omega}_{n+1})$ using the midpoint angular velocity in eq. (19) and the midpoint angular acceleration in the $\boldsymbol{\omega}$ update. Evaluating the inertial term $\boldsymbol{\omega} \times (\mathbf{l}\boldsymbol{\omega})$ at the midpoint gives the method its second-order accuracy on the gyroscopic dynamics.

6 Coupled time integration: the impulse scheme

The fluid load depends on the end-of-step body state, and the end-of-step state depends on the load: the coupling is implicit. We use an implicit-midpoint scheme in which the force and torque are formed from *differences of the state-function impulse* between the step endpoints, which is what makes the discrete scheme conserve momentum and energy.

6.1 Step equations

Over a step $t_n \rightarrow t_{n+1} = t_n + \Delta t$ let subscripts $n, n+1$ denote endpoint quantities. The impulses $\mathbf{L}^n, \boldsymbol{\Lambda}^n$ are evaluated once at the start state. The end velocity $\mathbf{v}^{n+1}$, end orientation, and step torque are the unknowns of a coupled fixed point. For each body i the residual equations are the implicit-midpoint momentum balance and the transported torque,

$$\mathbf{r}_i^{\text{lin}} = m_i(\mathbf{v}_i^{n+1} - \mathbf{v}_i^n) - (\mathbf{L}_i^{n+1} - \mathbf{L}_i^n) = \mathbf{0}, \quad (20)$$

$$\mathbf{N}_i = \frac{\boldsymbol{\Lambda}_i^{n+1} - \boldsymbol{\Lambda}_i^n}{\Delta t} - \bar{\mathbf{v}}_i \times \bar{\mathbf{p}}_i, \quad (21)$$

with midpoint averages $\bar{\mathbf{v}}_i = \frac{1}{2}(\mathbf{v}_i^n + \mathbf{v}_i^{n+1})$ and $\bar{\mathbf{p}}_i = \frac{1}{2}[(m_i\mathbf{v}_i^n - \mathbf{L}_i^n) + (m_i\mathbf{v}_i^{n+1} - \mathbf{L}_i^{n+1})]$. Positions advance with the midpoint velocity, $\mathbf{c}_i^{n+1} = \mathbf{c}_i^n + \bar{\mathbf{v}}_i\Delta t$, and the orientation advances by the PCDM update driven by $\mathbf{N}_i$. Equation eq. (21) is the discrete form of eq. (13); the transport term uses the per-body conserved momentum $\bar{\mathbf{p}}_i$, consistent with eq. (20) which enforces $m_i\Delta\mathbf{v}_i = \Delta\mathbf{L}_i$ (hence $\Delta\mathbf{p}_i = 0$) at convergence.

6.2 Fixed-point solution

Each evaluation of eq. (20)–eq. (21) requires a BEM solve to obtain $\mathbf{L}^{n+1}, \boldsymbol{\Lambda}^{n+1}$ at the trial end state. The fixed point couples the translational velocity (through $\mathbf{r}^{\text{lin}}$) and the torque (through the orientation it drives). The base iteration is:

- a *preconditioned quasi-Newton* step for the velocity, $\mathbf{v}_i \leftarrow \mathbf{v}_i - \mathbf{P}_i^{-1}\mathbf{r}_i^{\text{lin}}$ with the body-frame added-mass preconditioner $\mathbf{P}_i = m_i\mathbf{I} + \mathbf{M}_{f,i}$, which is exact for an isolated body and a good approximation otherwise; and
- a functional update of the torque from eq. (21).

Stacking the per-body iterate $\mathbf{w} = (\mathbf{v}^{n+1}, \mathbf{N})$ and the base-map correction $\mathbf{f}(\mathbf{w}) = \mathbf{w}' - \mathbf{w}$, the fixed point $\mathbf{f}(\mathbf{w}) = \mathbf{0}$ is solved to a tolerance $\varepsilon = 10^{-9}(\|\mathbf{v}^n\| + 1)$.

6.3 Anderson acceleration

The torque component contracts slowly (a near-unit linear rate), so plain functional iteration needs many sweeps—each a full BEM solve. We accelerate with Anderson’s method of depth

Algorithm 1 One coupled time step (impulse scheme, body loop implicit)

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1: evaluate start-state impulses  $\mathbf{L}^n, \mathbf{\Lambda}^n$  and  $T_f^n$  (one BEM solve)
2: initialise  $\mathbf{v}^{n+1} \leftarrow \mathbf{v}^n$ ,  $\mathbf{N} \leftarrow \mathbf{0}$ , empty Anderson history
3: for  $k = 0, 1, 2, \dots$  until converged or  $k = k_{\max}$  do
4:   advance orientation by PCDM with current  $\mathbf{N}$ ; set  $\mathbf{c}^{n+1}$  from  $\bar{\mathbf{v}}$ 
5:   solve BEM at the trial end state  $\rightarrow \mathbf{L}^{n+1}, \mathbf{\Lambda}^{n+1}$  ▷ warm-started GMRES
6:   form  $\mathbf{r}^{\text{lin}}$  eq. (20) and  $\mathbf{N}$  eq. (21)
7:   if  $\|\mathbf{r}^{\text{lin}}\| < \varepsilon$  and torque change  $< \varepsilon$  then break
8:   end if
9:   quasi-Newton velocity step  $\mathbf{P}^{-1}\mathbf{r}^{\text{lin}}$ ; Anderson-update  $(\mathbf{v}^{n+1}, \mathbf{N})$  eq. (22)
10: end for
11: commit  $\mathbf{v}^{n+1}$ ,  $\mathbf{c}^{n+1}$ , orientation; record  $T = T_f + T_s$ ,  $\mathbf{P}$ ,  $\mathbf{H}$ 

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m . Storing the last $m+1$ iterate/correction pairs and forming the difference matrices $\Delta\mathbf{F} = [\Delta\mathbf{f}_{k-1}, \dots, \Delta\mathbf{f}_{k-m}]$, $\Delta\mathbf{W} = [\Delta\mathbf{w}_{k-1}, \dots, \Delta\mathbf{w}_{k-m}]$, the next iterate is

$$\gamma = \arg \min_{\gamma} \|\mathbf{f}_k - \Delta\mathbf{F} \gamma\|_2, \quad \mathbf{w}_{k+1} = \mathbf{w}_k + \mathbf{f}_k - (\Delta\mathbf{W} + \Delta\mathbf{F}) \gamma, \quad (22)$$

where the small least-squares problem is solved through regularised normal equations. Anderson acceleration changes only the path to the fixed point, not the fixed point itself: the converged state still satisfies eq. (20)–eq. (21) to the same tolerance ε . In practice it roughly halves the number of BEM solves per step.

7 Linear algebra and performance

The dominant cost is the repeated solution of eq. (7). Several structural facts are exploited.

Cached self-blocks (block preconditioner). The diagonal block of $\mathbf{A}$ for body i (its self-influence) is invariant under rigid-body motion, so it is assembled and LU-factorised once. For a single body this is the whole matrix and the step reduces to a back-substitution. For N bodies the cached self-block factorisations form a block-diagonal preconditioner $\mathbf{M}^{-1}$, and only the smooth off-diagonal interaction blocks are reassembled each solve.

Preconditioned GMRES. Because eq. (5) is second kind, the preconditioned system $\mathbf{M}^{-1}\mathbf{A}\phi = \mathbf{M}^{-1}\mathbf{b}$ has clustered spectrum and GMRES converges in a handful of iterations. The matrix–vector product applies the freshly assembled interaction blocks plus the cached self-blocks; the preconditioner applies the cached LU factors.

Warm starting. Consecutive solves—across fixed-point iterations within a step and across steps—have nearly identical potentials, so each GMRES call is initialised from the previous solution. This typically reduces the iteration count to one to three.

Geometry caching. The reference subdivided mesh and its connectivity depend only on n_{div} and are memoised; each solve applies only the rigid-body transform to the cached body-frame nodes.

Complexity. With n surface nodes the assembly and each GMRES matrix–vector product are $\mathcal{O}(n^2)$. The per-step cost is (solves per step) $\times \mathcal{O}(n^2)$; warm starting cuts the GMRES factor and Anderson acceleration cuts the solves-per-step factor, while the analytic added-mass preconditioner keeps the implicit velocity solve robust as $\rho_s/\rho_f \rightarrow 0$.

Table 1: Temporal convergence at $n_{\text{div}} = 3$, $T = 8$. Self-error is versus the $\Delta t_{\text{ref}} = 0.0125$ run; the analytic error plateaus on the spatial floor.

Δt	self-error	ratio	error vs analytic	max energy drift
0.40	4.40×10^{-3}	—	0.1604	1.043%
0.20	1.09×10^{-3}	4.0	0.1621	0.267%
0.10	2.73×10^{-4}	4.0	0.1627	0.071%
0.05	6.54×10^{-5}	4.2	0.1628	0.019%
fitted order $p = 2.02$				

8 Convergence: single body

The method has two independent discretisation parameters—the time step Δt and the mesh refinement n_{div} —and we verify each in turn on the cleanest possible test: a single translating and rotating triaxial ellipsoid ($a:b:c = 1:0.8:0.6$, equivalent radius 1, $\rho_s = \rho_f$, initial $\mathbf{v} = (-1, 0, 0)$, $\boldsymbol{\omega} = (1, 1, 0)$). For one body the fluid load is a pure added-mass effect, so the exact motion is the force-free Kirchhoff system with the constant analytic added-mass and added-inertia tensors of section 4. Integrating that low-dimensional ODE with a finely sub-stepped Runge–Kutta scheme gives an essentially exact reference trajectory $\mathbf{x}_{\text{exact}}(t)$ and an exact initial fluid kinetic energy $T_f^{\text{exact}}(0) = 0.7792$, against which the BEM solver is measured. All errors below are reported at the final time $T = 8$.

8.1 Temporal convergence

We fix the mesh at $n_{\text{div}} = 3$ and refine $\Delta t \in \{0.4, 0.2, 0.1, 0.05\}$. The temporal order is isolated by *self-convergence*: the error of the centre position $\mathbf{x}(T)$ against the same-mesh run at the finest step $\Delta t_{\text{ref}} = 0.0125$, which removes the spatial-discretisation contribution. Table 1 and the left panel of fig. 1 show a clean second-order rate,

$$\|\mathbf{x}(T; \Delta t) - \mathbf{x}(T; \Delta t_{\text{ref}})\| = \mathcal{O}(\Delta t^p), \quad p \approx 2.0, \quad (23)$$

consistent with the implicit-midpoint impulse update for translation and the predictor–corrector PCDM update for rotation, both formally second order. The maximum total-energy drift over the run falls at the same rate (1.04% $\rightarrow$ 0.27% $\rightarrow$ 0.071% $\rightarrow$ 0.019%, a factor ≈ 3.8 per halving of Δt), confirming that the energy error of the impulse scheme is a genuine $\mathcal{O}(\Delta t^2)$ truncation effect, not a secular instability.

The error measured against the *analytic* reference (dashed line, left panel) behaves differently: it decreases with Δt only until it reaches a floor of ≈ 0.16 set by the spatial discretisation at $n_{\text{div}} = 3$, then stops improving. This is the practically important observation: once Δt is small enough that the temporal error drops below the spatial floor, further reducing Δt buys nothing.

8.2 Grid (spatial) convergence

We now fix a fine step $\Delta t = 0.025$ (temporal error negligible) and refine the mesh, $n_{\text{div}} \in \{1, 2, 3, 4\}$, i.e. $66 \rightarrow 258 \rightarrow 1026 \rightarrow 4098$ nodes with nominal element size $h \propto 2^{-n_{\text{div}}}$. Two errors against the analytic reference are reported: the fluid kinetic energy at $t = 0$ (a pure boundary-operator metric that involves no time integration) and the end-of-run centre displacement. Both, in table 2 and the right panel of fig. 1, halve under each mesh refinement, i.e. first order in the element size,

$$\frac{|T_f^{\text{BEM}}(0) - T_f^{\text{exact}}(0)|}{T_f^{\text{exact}}(0)} = \mathcal{O}(h^q), \quad q \approx 1.0\text{--}1.25. \quad (24)$$

Table 2: Grid convergence at $\Delta t = 0.025$, $T = 8$. N is the node count; $h \propto 2^{-n_{\text{div}}}$. Both errors are relative to the analytic reference.

n_{div}	N	h	fluid-KE error	ratio	displacement error
1	66	0.500	0.661	—	1.039
2	258	0.250	0.319	2.1	0.375
3	1026	0.125	0.153	2.1	0.163
4	4098	0.0625	0.0746	2.1	0.0765
fitted orders: $q_{\text{KE}} = 1.05$, $q_{\text{traj}} = 1.25$					

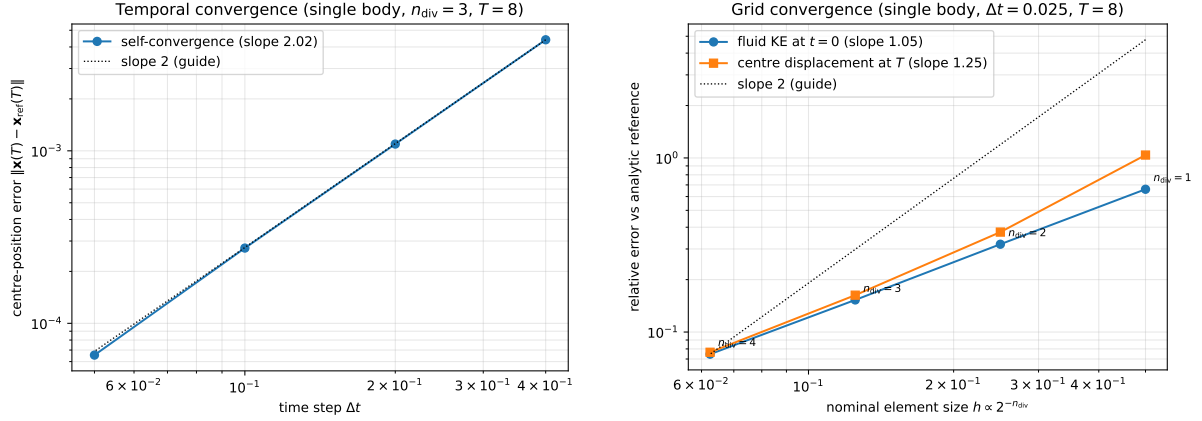


Figure 1: Single-body convergence. *Left*: temporal convergence at $n_{\text{div}} = 3$ by self-convergence (centre-position error versus the $\Delta t = 0.0125$ run); the slope is a clean second order. *Right*: grid convergence at $\Delta t = 0.025$ —the fluid kinetic energy at $t = 0$ and the end-of-run displacement both converge at first order in the element size h .

The first-order spatial rate—rather than the higher order one might hope for from quadratic six-node elements—reflects the collocation discretisation with the present singular-integration and surface-representation choices, and it is the accuracy bottleneck of the whole method. This is the root cause of the plateau seen in the temporal study and, more broadly, of the observation that long-time energy drift is essentially independent of Δt : the floor is spatial. To halve the error of a simulation one must refine the mesh (which costs $\mathcal{O}(h^{-2})$ nodes and $\mathcal{O}(h^{-4})$ work), not the time step. Conversely, Δt should be chosen only fine enough that the $\mathcal{O}(\Delta t^2)$ temporal error sits at or just below the $\mathcal{O}(h)$ spatial floor; at $n_{\text{div}} = 3$ the floor is ≈ 0.16 in centre position, which the modest step $\Delta t = 0.1$ (self-error 3×10^{-4}) already clears by three orders of magnitude.

8.3 Long-time energy conservation

Figure 2 tracks the total-energy drift over a long run ($T = 50$) as each parameter is varied in isolation. Refining the time step at a fixed mesh (left panel, $n_{\text{div}} = 2$) reduces the drift monotonically ($0.21\% \rightarrow 0.068\% \rightarrow 0.033\%$ for $\Delta t = 0.2, 0.1, 0.05$), the expected temporal behaviour. Refining the *mesh* at a fixed (coarse) step (right panel, $\Delta t = 0.2$) does *not* improve conservation—the drift in fact rises slightly ($0.21\% \rightarrow 0.27\% \rightarrow 0.30\%$ for $n_{\text{div}} = 2, 3, 4$). This is not a contradiction: at $\Delta t = 0.2$ the error is temporally dominated, and a finer mesh resolves slightly stiffer added-mass dynamics that the fixed step integrates marginally less accurately. The lesson is that energy conservation is governed by the *balance* of the $\mathcal{O}(\Delta t^2)$ temporal and $\mathcal{O}(h)$ spatial errors: both must be refined together, and there is no benefit to reducing one far below the other.

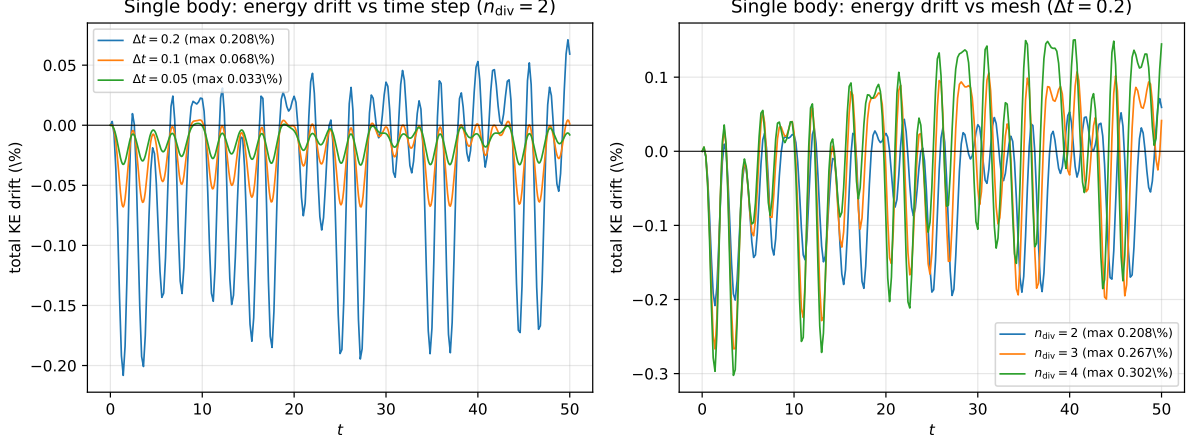


Figure 2: Single-body total-energy drift over $T = 50$. *Left*: time-step refinement at $n_{\text{div}} = 2$ (drift falls with Δt). *Right*: mesh refinement at $\Delta t = 0.2$ (drift does not fall—the error is temporally limited at this step).

8.4 Orientation tracking across density ratios

The scalar metrics above are complemented by a direct visual comparison of the *orientation marker*—the body x -axis expressed in the lab frame, a unit vector that traces the body’s tumbling. Figure 3 overlays the marker components from the BEM at $n_{\text{div}} = 2$ and 3 on the analytic reference for two representative members of a ten-run ensemble (randomised initial orientations and spin axes) spanning the fluid-to-solid kinetic-energy ratio. For a heavy body at energy ratio ≈ 1 (top row) the coupling is weak and both meshes track the exact tumbling essentially perfectly over the whole window. For a light body at energy ratio ≈ 20 (bottom row)—where the added-mass torque dominates the dynamics—the coarse $n_{\text{div}} = 2$ marker accumulates a visible phase error, while $n_{\text{div}} = 3$ recovers the exact orientation far more closely. This is the same first-order spatial behaviour of section 8 seen directly in the configuration variables: the hardest, added-mass-dominated regime is the most mesh-sensitive.

9 Convergence: multiple bodies

For more than one body the added mass is configuration dependent, so there is no closed-form reference. We therefore verify convergence by *self-comparison* against a finely resolved run, on a three-body configuration of three interacting triaxial ellipsoids placed on a ~ 5 -radius triangle with orbiting velocities and distinct spins.

9.1 Temporal convergence

Fixing $n_{\text{div}} = 2$ and the neutral density ratio $\rho_s = \rho_f$, we refine $\Delta t \in \{0.2, 0.1, 0.05\}$ and measure the error of the combined three-body centre configuration $\{\mathbf{c}_i(T)\}$ at $T = 25$ against a reference run at $\Delta t_{\text{ref}} = 0.025$. The rate is again second order (table 3, fig. 4),

$$\left(\sum_i \|\mathbf{c}_i(T; \Delta t) - \mathbf{c}_i(T; \Delta t_{\text{ref}})\|^2 \right)^{1/2} = \mathcal{O}(\Delta t^p), \quad p \approx 2.2, \quad (25)$$

confirming that the implicit impulse / PCDM time integration retains its formal order under the full multi-body fluid coupling, including the reference-point transport term eq. (13).

Table 3: Three-body temporal self-convergence ($\rho_s = \rho_f$, $n_{\text{div}} = 2$, $T = 25$); error versus the $\Delta t_{\text{ref}} = 0.025$ run, summed over the three centres.

Δt	self-error	ratio	max energy drift
0.20	1.43×10^{-3}	—	0.463%
0.10	3.36×10^{-4}	4.3	0.401%
0.05	6.67×10^{-5}	5.0	0.389%
fitted order $p = 2.21$			

9.2 Energy conservation and the spatial floor

The energy-drift column of table 3 shows the same spatial-floor behaviour as the single body: as Δt falls the drift flattens onto $\approx 0.386\%$ (the value at $\Delta t = 0.025$), so the residual is spatial, not temporal. Figure 5 makes this explicit—the left panel shows the drift history collapsing onto a common envelope as Δt is refined, and the right panel plots the maximum drift against Δt for $n_{\text{div}} = 2$ and 3 (at $\rho_s = \rho_f$): both curves are nearly flat, and lowering the floor requires a finer mesh, not a finer step. Refining the mesh from $n_{\text{div}} = 2$ to 3 reduces the drift modestly ($0.40\% \rightarrow 0.37\%$ at $\Delta t = 0.1$) but shifts the trajectory substantially—the combined centre configuration moves by $\|\Delta \mathbf{c}\| \approx 0.76$ between the two meshes—which is fully consistent with the first-order spatial accuracy established for the single body: the mesh is the dominant error, and the coarse meshes are far from the spatial asymptote.

A full multi-body *grid*-convergence order is not reported here: the next mesh level ($n_{\text{div}} = 4$) for three bodies is an $\mathcal{O}(n^2)$ -per-solve cost roughly $16\times$ that of $n_{\text{div}} = 3$ and is impractical for a refinement sweep, and the lightest-density fine-mesh case is past the stability limit noted below. The single-body study of section 8 establishes the spatial order rigorously against the analytic reference; the multi-body data are consistent with it.

10 Conservation and verification

Because the loads are exact differences of the state functions eq. (8)–eq. (9) and the torque carries the transport term eq. (13), the scheme conserves the total linear and angular momentum eq. (14) to the fixed-point tolerance, and the total kinetic energy to the temporal truncation error of the integrator. The natural verification metrics are the relative drift of T and the span of $\mathbf{P}$ and $\mathbf{H}$ over a run. Linear momentum is conserved essentially to round-off; angular-momentum conservation is the sensitive diagnostic that exposes a missing or inconsistent transport term; and the energy drift, once momentum is conserved, is dominated by the spatial (mesh/quadrature) discretisation rather than the time step.

Stability limit. The translational coupling is treated implicitly and is robust, but the rotational PCDM update is explicit in the gyroscopic and added-inertia terms. For very light bodies on fine meshes (large added-to-solid inertia ratio) this explicit angular treatment can exceed its stability limit and the orientation blows up, independently of the time step; such configurations lie outside the method’s stable regime and must be flagged rather than trusted.

Nomenclature

Symbol	Meaning
ϕ	velocity potential, $\mathbf{u} = \nabla\phi$
$\partial_n\phi$	surface-normal derivative of ϕ (Neumann data)
$\mathbf{A} = \mathbf{D} - \frac{1}{2}\mathbf{I}$	discrete BEM influence matrix
$\mathbf{S}, \mathbf{D}$	discrete single- / double-layer operators
$\mathbf{L}_i, \mathbf{\Lambda}_i$	fluid linear / angular impulse of body i
T_f, T_s	fluid / solid kinetic energy
$\mathbf{p}_i, \mathbf{H}_i$	conserved per-body linear / angular momentum
$\mathbf{M}_f, \mathbf{I}_f$	added-mass / added-inertia tensors
$\mathbf{P}_i = m_i\mathbf{I} + \mathbf{M}_{f,i}$	velocity-solve preconditioner
$q_i, \boldsymbol{\omega}_i$	orientation quaternion / angular velocity of body i
$n_{\text{div}}, n_{\text{pts}}$	mesh subdivision level / node count
m_{int}, n_q	triangle / Gauss–Legendre quadrature orders
$\Delta t, \varepsilon$	time step / fixed-point tolerance

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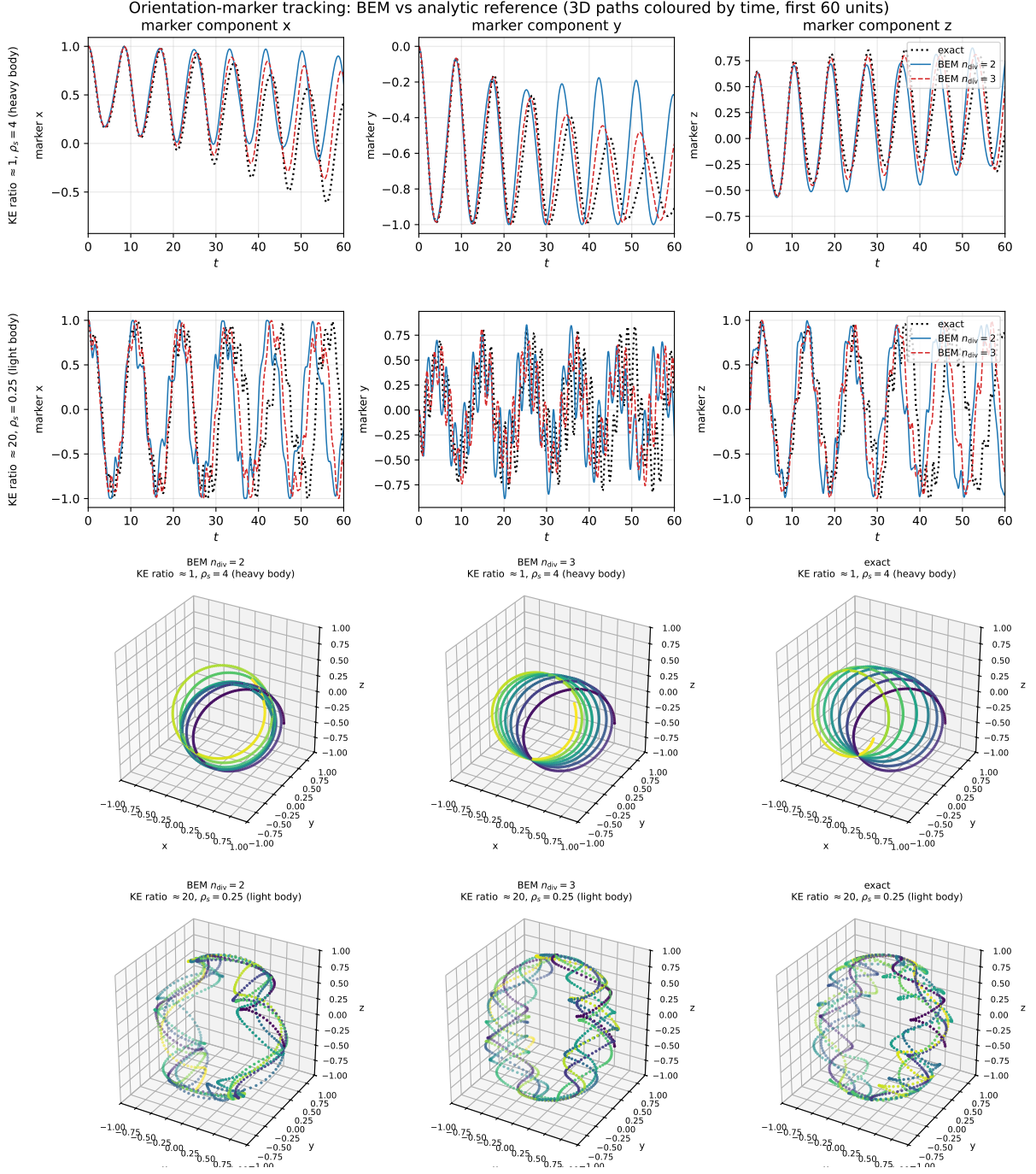


Figure 3: Orientation-marker tracking (body x -axis in the lab frame) for two single-body runs. *Rows 1–2:* the three marker components versus time, overlaying BEM $n_{\text{div}} = 2$ (blue) and $n_{\text{div}} = 3$ (red dashed) on the analytic reference (black dotted)—energy ratio ≈ 1 , $\rho_s = 4$ (tracked at both meshes) and energy ratio ≈ 20 , $\rho_s = 0.25$ (coarse mesh drifts, finer mesh recovers the tumbling). *Rows 3–4:* the marker as a 3D trajectory of the unit vector on the sphere, with a separate plot for $n_{\text{div}} = 2$, $n_{\text{div}} = 3$ and the exact reference—row 3 for the heavy-body case, row 4 for the light-body case—coloured by time. For the light body the $n_{\text{div}} = 2$ path fills a visibly different region of the sphere than the reference, whereas $n_{\text{div}} = 3$ closely matches it.

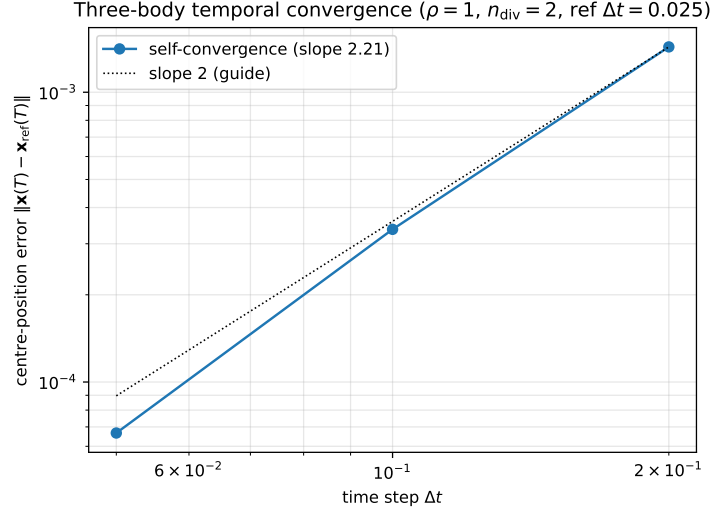


Figure 4: Three-body temporal self-convergence ($\rho_s = \rho_f$, $n_{\text{div}} = 2$, $T = 25$): the combined centre-configuration error versus the $\Delta t = 0.025$ reference is second order (slope ≈ 2).

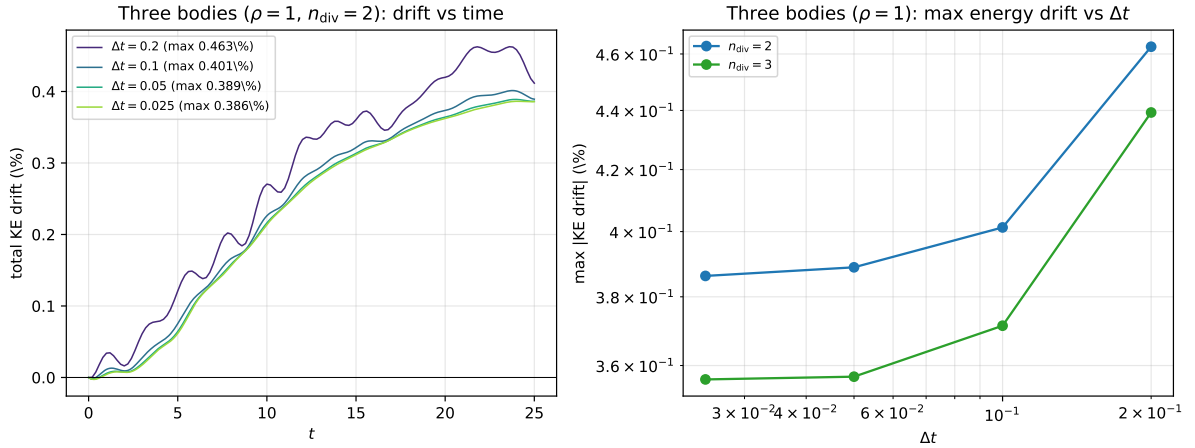


Figure 5: Three-body energy conservation ($\rho_s = \rho_f$). *Left*: drift history over $T = 25$ for the time-step sweep at $n_{\text{div}} = 2$. *Right*: maximum total-energy drift versus Δt for $n_{\text{div}} = 2$ and 3; both curves are nearly flat—pinned on the spatial floor—so reducing Δt alone does not improve conservation, whereas refining the mesh lowers the floor.